Assignment 1

1. (b) Given: Antenna gain $G_t = 15$ dBi

$= 10^{15/10} = 31.6$

Transmitted power $P_t = 20W$

Distance $d = 50m$

Power density at a distance 'd' is given by-

$$P_d = \frac{P_t G_t}{4\pi d^2} \text{ Watt}/m^2$$

$$= \frac{20 \times 31.6}{4\pi \times (50)^2} = 20.1 \text{ mW}/m^2$$

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2. (c) Given: Transmitting antenna gain = 18 dBi

$G_t = 10^{1.8} \simeq 63.1$

Receiving antenna gain = 15 dBi

$G_r = 10^{1.5} = 31.6$

Transmitted power $P_t = 20W$

Distance 'd' = 2 km = 2000 m

Frequency $f = 5.8$ GHz

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5.8 \times 10^9} = 0.0517 \text{ m}$$

Received power is given by $P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$ Watt

$$P_r = 20 \times 63.1 \times 31.6 \times \left(\frac{0.0517}{4\pi \times 2000}\right)^2 = 0.169 \ \mu W$$

$$= 20 \log\left(\frac{P_r}{1mW}\right) \text{ dBm}$$

$$\boxed{P_r = -37.7 \text{ dBm}}$$

3.(d) The output power of low noise amplifiers is less than 10 dBm.

4.(b) The carrier frequency of GPS receiver module is 1575 MHz.

5.(d) The GPS modules can be used in -

- Vehicle tracking
- Remote monitoring
- Location identification
- People tracking etc.

6.(a) The frequency range of AM radio is from 530 KHz – 1620 KHz.

7.(b) In USA, the maximum allowed SAR value is 1.6 W/Kg for 6 minutes of usage.

8.(a) The extremely high radiation zone of a cell tower is at a distance less than 50m.

9.(c) In 2011, WHO has classified cell tower radiation as "Class 2B - possible carcinogen" to human.

10.(d) The excessive use of cell phone over a long period of time may cause -

- Hearing problem
- Increase in cancer risk
- Irreversible infertility
- DNA damage
- Immune system degradation

etc.